Reference Manual for mohelper

thevicraft

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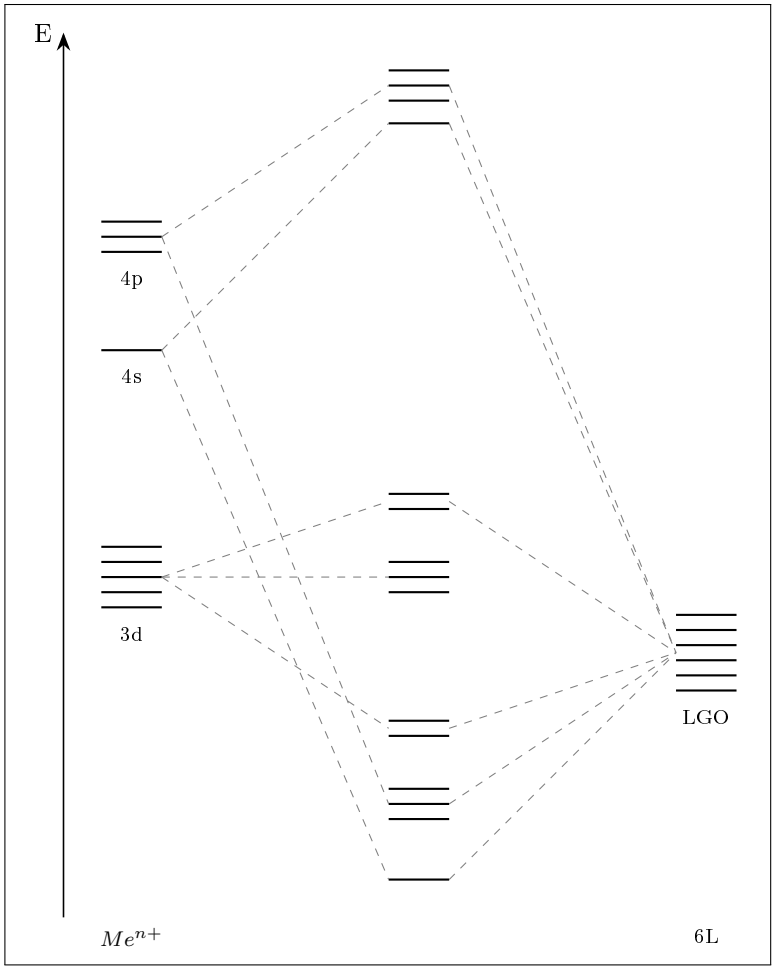

Figure 1: Octahedral Complex

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1 Introduction

Stop fighting with pure TikZ. mohelper is a LaTeX package designed to create professional Molecular Orbital (MO) diagrams with minimal code. It handles the positioning, electron population, and dashed connection lines automatically, so you can focus on the chemistry, not the coordinates. Have a look:

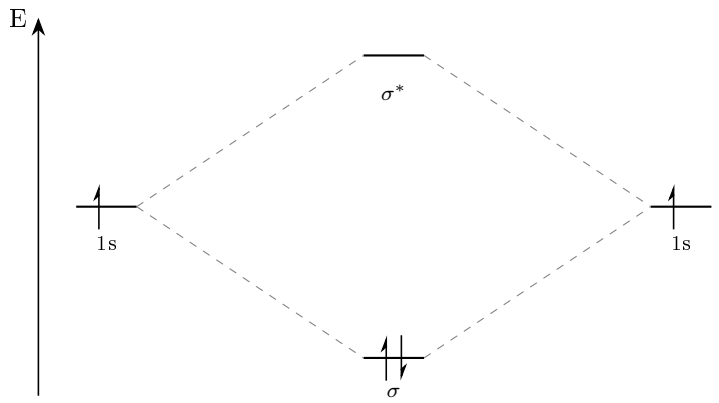

Figure 2: Hydrogen Molecule

```
\begin{mohelper}
    \def\columndistance{3}
    \def\diagrammode{normal}
    \degenerate{0}{1s/up}{left}
    \degenerate{-2}{$\sigma$/pair}{middle}
    \degenerate{2}{$\sigma^*$/empty}{middle}
    \degenerate{0}{1s/up}{right}
    \connectorbital{0}{1}{-2}{1}{left}
    \connectorbital{0}{1}{2}{1}{left}
    \connectorbital{-2}{1}{0}{1}{right}
    \connectorbital{2}{1}{0}{1}{right}
    \addenergyscale{0}
\end{mohelper}
```

2 Quick Start

To use the diagrams generated by this tool in your LaTeX project, follow these steps:

2.1 Include the Package

Download the package and include it into your project. To do so download the `mohelper.sty` file from

1. Releases (recommended, stable)
2. Repository (only if you know what you are doing)

and paste the file into your project's root directory. Ensure you have the mohelper package included in your preamble:

```
\input{mohelper.sty}
```


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If you put `mohelper.sty` into a subdirectory, you must specify its path. Therefore, have this example shown:

```
\input{./lib/mohelper.sty}
```

2.2 Configure your Diagram

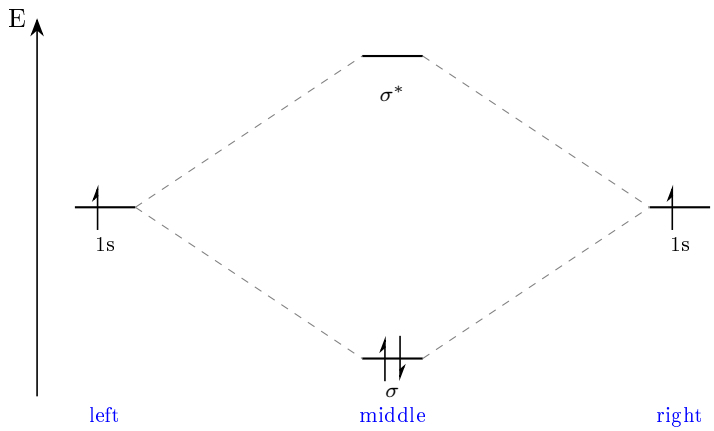
Open the Web Helper and set up your orbitals, energies, and labels.

2.3 Copy and Paste

Copy the generated LaTeX code and paste it into your `figure` environment or document body.

3 Basic Usage

The heart of the package is the `mohelper` environment. It follows a three-column logic: **Left (AO)** | **Middle (MO)** | **Right (AO)**.



4 Macro Reference

4.1 The Environment

```
\begin{mohelper}[options]
    ...
\end{mohelper}
```

Wraps a `tikzpicture`. You can pass standard TikZ options like `[scale=0.8]`.

4.2 Defining Orbitals

```
\degenerate{energy}{label1/spin1, label2/spin2, ...}{side}
```

The `\degenerate` command is your Swiss Army knife. It automatically calculates the centering for multiple orbitals.

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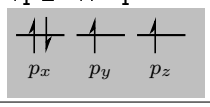
Parameter	Description	Example
#1	**Y-Level** (Energy)	`-2.5`
#2	**Label/Spin Array** (Format: `label1/spin1, label2/spin2, ...`)	`$p_x$/pair, $p_y$/up, $p_z$/up` 
#3	Placement	`left`, `middle` or `right`

Supported Spins: `up`, `down`, `pair` (any other value will result in an empty orbital)

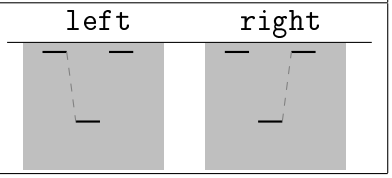
```
\orbital{x}{y}{label}{spin}
```

`\orbital` manually draws a single orbital at the specified coordinates. For standard orbital diagrams it should not be necessary to use this macro. Only use `\degenerate`.

Parameter	Description	Example
#1	**x** coordinate	`4.2`
#2	**y** coordinate (energy)	2.0
#3	Label	`1s`
#4	Spin	`up`, `down` or `pair` (any other value will result in an empty orbital)

4.3 Connecting Orbitals

```
\connectorbital{left-E}{left-deg}{right-E}{right-deg}{side}
```

Parameter	Description	Example
#1	**Energy** of **left** orbital	`-5.3`
#2	**Degeneration** of **left** orbital	`1`
#3	**Energy** of **right** orbital	`2.5`
#4	**Degeneration** of **right** orbital	`3`
#5	**Placement** of the connection line	

By specifying a side (`left` or `right`) in `#5` the connection line is drawn between either

- `left`: left AO and MO
- `right`: MO and right AO

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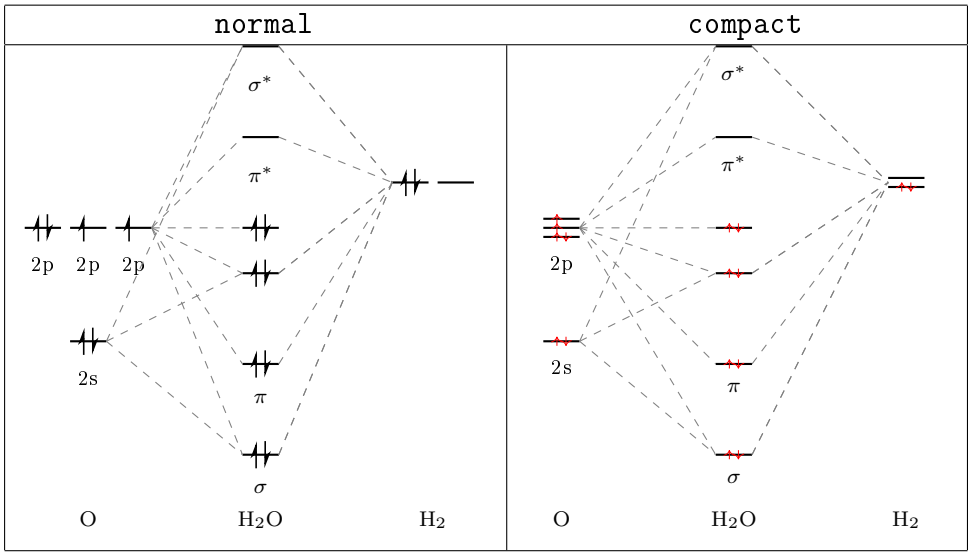
4.4 Customization

You can tweak the look for each diagram by redefining these values:

- `\def\columndistance{value}`: Set `value` to a number. It defines space between the AO und MO columns.
- `\def\diagrammode{compact}`: Switches to vertical stacking for degenerate orbitals (perfect for large systems).



- `\def\electroncolor{color}`: Set `color` to a valid color value. Changes the electron color to a specified one. The redefining has to stand before the relevant orbital is created using `\degenerate`.

4.5 Miscellaneous

```
\degeneratelabel{energy}{deg}{side}{align}{text}
```

Draws a label for a group of degenerated orbitals. This is an alternate macro if the label is not specified in `\degenerate`. Furthermore, the label is intended to be describing all degenerated orbitals of concern. Here, you can specify the text position, very helpful if you want the label next to the degenerated orbitals (see param `#4`).

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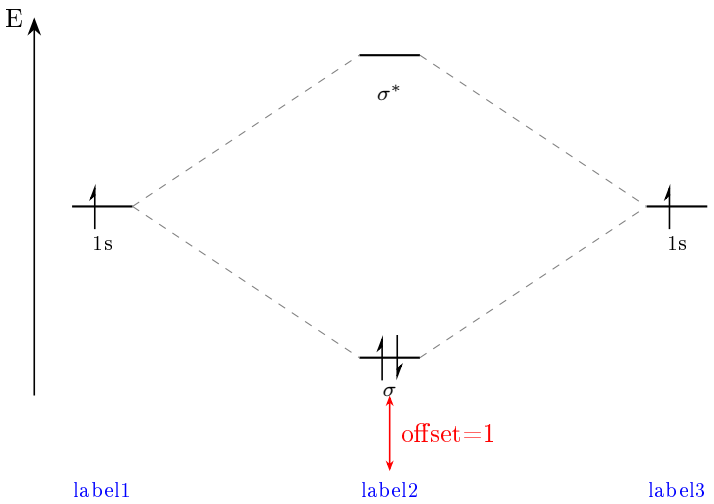
Parameter	Description	Example
#1	**Energy** of orbital	`-13.6`
#2	**Degeneration** of orbital	`1`
#3	**Column** of orbital	`left` left AO `middle` MO `right` right AO
#4	**Text Position** relative to orbital	`left` $1a_{1g}$ `right` $1a_{1g}$ `bottom` $1a_{1g}$ `top` $1a_{1g}$
#5	**Text Content**	

```
\addenergyscale{offset}
```

Adds an energy scale. `offset` (number) specifies the additional horizontal offset from the diagram. As a default, just set it to `0`. This macro has to be executed after each orbital is drawn. Otherwise the scale's size and position will be calculated incorrect.

```
\addlabel{label1, label2, label3}{offset}
```

Gives all three columns a label (left AOs, MOs, right AOs). `offset` (number) specifies the additional vertical offset from the diagram. As a default, just set it to `0`. This macro has to be executed after each orbital is drawn. Otherwise the vertical position will be calculated incorrect.



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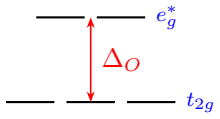
```
\showenergydifference{energy1, energy2}{side}{offset}{text}
```

Draws a double-sided red arrow pointing out an energy difference. `offset` (number) specifies an additional horizontal offset. As a default, just set it to `0` which will center it.

Parameter	Description	Example
#1	**Energy Array** (Format: `energy1, energy2`)	`2.0, 4.0`
#2	**Column** of orbital	`2.0`
#3	**Horizontal offset**, moves the arrow along x-axis	`0` (centered)
#4	**Label**	`$\Delta_O$` (Δ_O)



5 Example

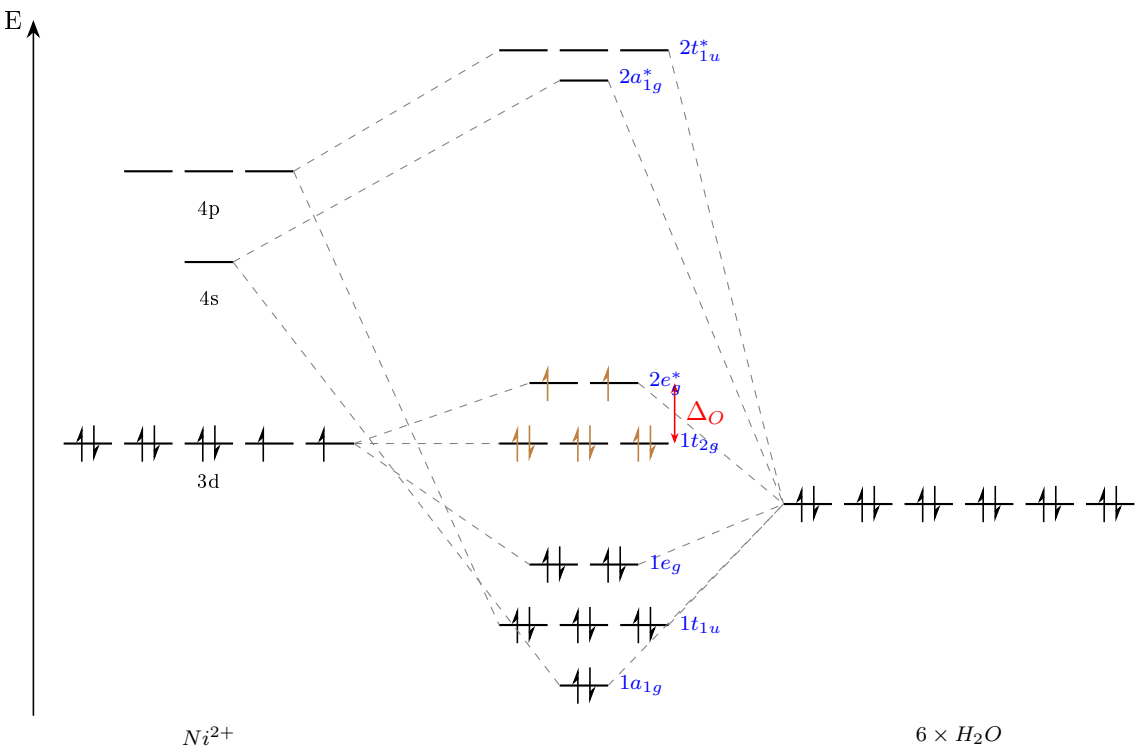


Figure 3: $[Ni(H_2O)_6]^{2+}$ complex

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```latex
\begin{mohelper}[scale=0.8] % scale it to 0.8 of original size to fit on
   page
    \def\columndistance{5.4} % set column distance to 5.4
    \def\diagrammode{normal} % draw in normal mode
    \degenerate{0}{/pair,/pair,3d/pair,/up,/up}{left}
    \degenerate{3}{4s/empty}{left}
    \degenerate{4.5}{/empty,4p/empty,/empty}{left}
    \def\electroncolor{brown} % change color for t2g
    \degenerate{0}{/pair,/pair,/pair}{middle}
    \def\electroncolor{black} % change color back for rest
    \degenerate{-4}{/pair}{middle}
    \degenerate{6}{/empty}{middle}
    \degenerate{-3}{/pair,/pair,/pair}{middle}
    \degenerate{6.5}{/empty,/empty,/empty}{middle}
    \degenerate{-2}{/pair,/pair}{middle}
    \def\electroncolor{brown} % change color for eg*
    \degenerate{1}{/up,/up}{middle}
    \def\electroncolor{black} % change color back for rest
    \degenerate{-1}{/pair,/pair,/pair,/pair,/pair,/pair}{right}
    \connectorbital{0}{5}{0}{3}{left}
    \connectorbital{0}{5}{-2}{2}{left}
    \connectorbital{0}{5}{1}{2}{left}
    \connectorbital{-2}{2}{-1}{6}{right}
    \connectorbital{1}{2}{-1}{6}{right}
    \connectorbital{3}{1}{-4}{1}{left}
    \connectorbital{3}{1}{6}{1}{left}
    \connectorbital{-4}{1}{-1}{6}{right}
    \connectorbital{6}{1}{-1}{6}{right}
    \connectorbital{4.5}{3}{-3}{3}{left}
    \connectorbital{4.5}{3}{6.5}{3}{left}
    \connectorbital{-3}{3}{-1}{6}{right}
    \connectorbital{6.5}{3}{-1}{6}{right}
    \addlabel{$Ni^{2+}$,,$6 \times H_2O$}{0} % draw labels
    \addenergyscale{0} % add the energy scale
    \showenergydifference{0,1}{middle}{1.5}{$\Delta_O$} % show ligand field
   splitting
    \degeneratelabel{6.5}{3}{middle}{right}{$\color{blue}{2t_{1u}^*}$}
    \degeneratelabel{6}{1}{middle}{right}{$\color{blue}{2a_{1g}^*}$}
    \degeneratelabel{1}{2}{middle}{right}{$\color{blue}{2e_g^*}$}
    \degeneratelabel{0}{3}{middle}{right}{$\color{blue}{1t_{2g}}$}
    \degeneratelabel{-2}{2}{middle}{right}{$\color{blue}{1e_g}$}
    \degeneratelabel{-3}{3}{middle}{right}{$\color{blue}{1t_{1u}}$}
    \degeneratelabel{-4}{1}{middle}{right}{$\color{blue}{1a_{1g}}$}
\end{mohelper}
```

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